

KarmaYogi Pragya

PROBLEM STATEMENT DETAILS

Problem Statement ID: SIH26101 (Ministry of Statistics & Programme Implementation — MoSPI)

Problem Statement Title: AI-enabled learning platform that identifies competency gaps, recommends personalized training via iGOT Karmayogi integration, and generates quizzes/MCQs from uploaded learning materials to strengthen capacity building in India's Official Statistical System.

Theme: Smart Education

PS Category: Software

Team ID: NA

Team Name: KarmaYogi Pragya

"Right skill. Right training. Right person. Right time."

Adaptive AI Diagnostics

Pragya AI Chat Copilot

Bilingual Voice Tutor

Supervisor Analytics Hub



OFFICIAL LEARNING JOURNEY

Simple 4-Step Flow

1



Quick Setup

Pick cadre, department, and language in 1 click.

2



3-Minute Test

10 smart MCQs with instant hints. If you get stuck.

3



Skill Scorecard

Instant report showing strengths and weak spots.

4



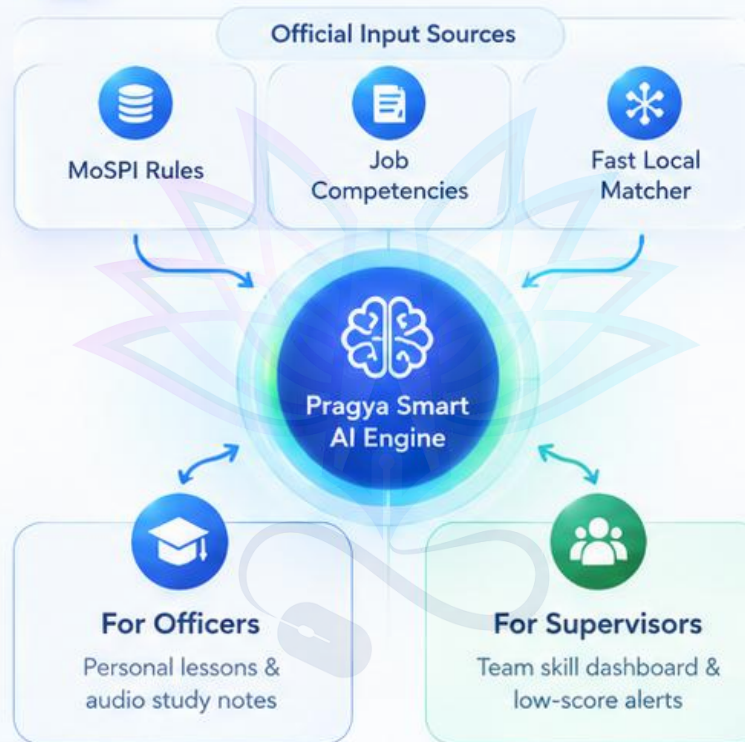
Learn & Grow

Bite-sized iGOT lessons with an interactive AI audio tutor.



PRAGYA SMART ENGINE

How It Works



WHY PRAGYA WINS

Old Way vs. Pragya



OLD:

Keyword Search
(Pick wrong courses)



PRAGYA:

Smart Match
(Finds exact skill gaps)



OLD:

Generic AI
(Hallucinates rules)



PRAGYA:

Official AI
(100% verified MoSPI rules)



OLD:

Zero Tracking
(Supervisors blind)



PRAGYA:

Live Alerts
(Notifies leads for scores <40%)



Zero Friction

Works directly on top of iGOT with no DB changes



100% Private

Runs locally on device with zero data leaks



Govt Approved

Fully compliant with National Training Policy



SYSTEM ARCHITECTURE

Edge-First Processing → Intelligent Routing → Secure Governance



Live Web App →

<https://karmayogi-pragya.onrender.com>

Scan for link



GitHub Repo →

<https://github.com/rudu2004/KarmaYogi-Pragya>

Scan for code



UI / FRONTEND

Vanilla JS • HTML5 • Web Speech API

Browser Client



FASTAPI BACKEND

Authentication • Telemetry • Workflows

API Layer



PRAGYA CORE ENGINE



Diagnostic
Engine



Vector Gap
Engine



AI Router



DATA + VECTOR LAYER



SQLite

Local Audit Store



Sentence-BERT

CPU Inference



MULTI-TIER AI FALLBACK CASCADE

Automatic failover with deterministic offline operation

01



Gemini Flash Lite

Fastest



02



Gemini Flash

Reliable



03



LLaMA 3.3 / Groq

High-Speed



04



GPT-OSS

Versatile



05



Free Fallback

Backup



06



Local Rules

Deterministic

OFFLINE



DPDP Act 2023 Compliant

Privacy • Data Protection • Governance



Containerized

Docker / Render



Zero iGOT Database Changes

Safe Integration



FOUR-PILLAR FEASIBILITY ANALYSIS

Proven. Practical. Scalable.



01

Technical

Ready & Scalable

- ✓ Zero-Schema Integration
- ✓ Local Processing
- ✓ Secure APIs



02

Operational

Simple & Efficient

- ✓ Minimal Setup
- ✓ 3-Min Diagnostics
- ✓ Auto Workflows



03

Institutional

Govt. Aligned

- ✓ MoSPI Alignment
- ✓ iGOT Compatible
- ✓ Standard Frameworks



04

Economic

Cost-Effective

- ✓ Zero Infrastructure Cost
- ✓ Open Source Stack
- ✓ High ROI



CHALLENGES & MITIGATION MATRIX

Addressing Risks for Reliable Deployment



RISK:

Staff Resistance & Exam Anxiety



SOLUTION:

Structured Training & Clear Onboarding

No formal appraisal linkage.



RISK:

Unstable Internet in Field Offices



SOLUTION:

Offline Local Fallback

Works without connectivity.



RISK:

AI Hallucinations in Statistics



SOLUTION:

Deterministic Formula Logic

Verified rules + calculations.



RISK:

Fear of Workplace Surveillance



SOLUTION:

Cohort-Level Telemetry Only

No individual tracking.



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IMPACT AND BENEFITS

Potential Impact on Civil Service Cadres | Benefits of the Solution (Governance & Economic)

3-DIMENSIONAL NATIONAL IMPACT



Individual Officer

Better skills • Higher confidence



Department / MoSPI

Higher competency • Better performance



National Governance

Data-driven decisions • Stronger India



QUANTIFIABLE IMPACT

Before vs. With Pragma



LEARNING & TESTING TIME

< 3 Min

With Pragma



2 – 4 Weeks

Traditional Way

↗ 95%+ Time Saved



DIAGNOSTIC ACCURACY

92%+

With Pragma



Keyword Search
Errors
Traditional Way

↗ High Precision



TARGET OFFICERS

3.3M+

Across India



Limited Reach
Traditional Way

↗ Full Coverage



DATABASE INFRASTRUCTURE

Zero

DB Changes



Required Schema Changes
Traditional Way

↗ Safe & Non-Invasive



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ACADEMIC & STATUTORY RESEARCH FOUNDATIONS

Official Mandates · Standards · Research Backing

01



MoSPI & NSSTA Operational Guidelines

Index number formulation · NSSTA core curriculum

02



Mission Karmayogi & DoPT FRAC Framework

CBC Training Policy · FRAC competency dictionaries

03



Semantic Vector Search

Reimers & Gurevych · EMNLP 2019 / Sentence-BERT

04



DPDP Act 2023

Ministry of Law & Justice · Purpose limitation & de-identified telemetry



VERIFIED PRODUCTION ARTIFACTS & DEMOS

Live Links · Source Code · Videos



Live Web Application

Containerized Render production portal (24/7 active)

<https://karmayogi-pragya.onrender.com>

URL



QR Code



Open-Source Repository

FastAPI backend, SQLite schema, Dockerfile & frontend

https://github.com/rudu2004/KarmaYogi_Pragya

URL



QR Code



Presentation Pitch Video

Complete slide-by-slide executive pitch & solution demo

URL



QR Code



Prototype Walkthrough Video

Live end-to-end demonstration & supervisor telemetry

URL



QR Code



100% Verifiable

Live deployed prototype and code accessible to all evaluators.



Evidence-Backed

Architecture directly anchored in published research and official gazettes.



Atmanirbhar GovTech

Fully open-source, zero proprietary locks, ready for institutional adoption.